DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM
August 2014
ALGEBRA (Ph.D. Version)

Instructions to the student

a. Answer all six questions; each will be assigned a grade from 0 to 10.

b. Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.

c. Keep scratch work on separate pages in the same booklet.

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1. Let G be a finite group and let N be a normal subgroup. Assume that $|N| = 3$ and that N is not contained in the center of G. Show that G contains a subgroup H with $[G : H] = 2$.

2. Let $n \geq 2$ and let A be a $n \times n$ integer matrix with 1's in every entry.
(a) Show that the characteristic polynomial of A is $x^{n-1}(x - n)$. (*Hint:* What is the rank of A ?)
(b) Find the Jordan canonical form of A. (*Hint:* $A^2 = nA$.)

3. **(a)** Prove that every element of $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ can be written in the form $x \otimes 1$ with $x \in \mathbb{Q}$. (*Note:* Recall that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ includes sums of elements of the form $a \otimes b$.)
(b) Show that the map $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathbb{Q}$ given by $a \otimes b \mapsto ab$ is an isomorphism of additive abelian groups.

4. Let R be a commutative Noetherian ring with 1, and let I be a nonzero ideal of R. Show that there are nonzero prime ideals $P_1, P_2, \ldots, P_n$, with $n \geq 1$, such that

$$P_1 P_2 \cdots P_n \subseteq I.$$

(*Note:* The product $P_1 P_2 \cdots P_n$ means the ideal generated by all products $a_1 a_2 \cdots a_n$ with $a_i \in P_i$ for all i.)

5. Let $K/\mathbb{Q}$ be a Galois extension of fields with $\mathrm{Gal}(K/\mathbb{Q}) \simeq A_5$, the simple group of order 60 consisting of even permutations of 5 objects.
(a) Let $f(x) \in \mathbb{Q}[x]$ be irreducible with $\deg f(x) \geq 2$. Show that if $f(x)$ has a root in K then $\deg f(x) \geq 5$.
(b) Show that there is an irreducible polynomial $g(x) \in \mathbb{Q}[x]$ of degree 20 such that $g(x)$ has a root in K.

6. The following is the character table of a group G:

	I	II	III	IV	V	VI
χ_1	1	1	1	1	1	1
χ_2	3	-1	1	0	$(-1+\sqrt{-7})/2$	$(-1-\sqrt{-7})/2$
χ_3	3	-1	1	0	$(-1-\sqrt{-7})/2$	$(-1+\sqrt{-7})/2$
χ_4	6	2	0	0	-1	-1
χ_5	7	-1	-1	1	0	0
χ_5	8	0	0	-1	1	1

The sizes of the conjugacy classes are as follows:

$$|I| = 1, \quad |II| = 21, \quad |III| = 42, \quad |IV| = 56, \quad |V| = 24, \quad |VI| = 24.$$

(a) Show that G is not a solvable group.
(b) For this part, you may use the fact that G is simple (that is, it has no non-trivial normal subgroups). Show that G is isomorphic to a subgroup of $\mathrm{GL}_3(\mathbb{C})$ (= invertible 3×3 complex matrices).
(c) How many subgroups of order 3 does G have? You may use the fact that in this group the elements of order 3 form a conjugacy class.